```
!pip install pandas spacy matplotlib emoji
get_ipython().system('python -m spacy download en_core_web_sm')
```

```
  Downloading emoji-2.15.0-py3-none-any.whl.metadata (5.7 kB)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.21.1)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.1)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (25.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.1)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.1
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.1
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->s
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->s
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4-
Requirement already satisfied: click>=8.0.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->spacy)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->spacy) (3.0.3)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5.0,
Downloading emoji-2.15.0-py3-none-any.whl (608 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 608.4/608.4 kB 10.2 MB/s eta 0:00:00
Installing collected packages: emoji
Successfully installed emoji-2.15.0
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 108.7 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
import pandas as pd
import spacy
import re
import emoji
from collections import Counter
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('/content/Tweets.csv')
```

```
data = pd.read_csv("Tweets.csv")
data = data[['text', 'airline_sentiment']]

print(data.head())
```

```
                                                text airline_sentiment
0                @VirginAmerica What @dhepburn said.           neutral
1  @VirginAmerica plus you've added commercials t...          positive
2  @VirginAmerica I didn't today... Must mean I n...           neutral
3  @VirginAmerica it's really aggressive to blast...          negative
4  @VirginAmerica and it's a really big bad thing...          negative
```

```
def clean_tweet(text):
    text = re.sub(r"http\S+", "", text)     # remove URLs
    text = re.sub(r"@\w+", "", text)        # remove mentions
    text = emoji.replace_emoji(text, "")    # remove emojis
    text = re.sub(r"\s+", " ", text)
    return text.strip()

data['clean_text'] = data['text'].apply(clean_tweet)

print(data['clean_text'].head())
```

```
0                                          What said.
1    plus you've added commercials to the experienc...
2    I didn't today... Must mean I need to take ano...
3    it's really aggressive to blast obnoxious "ent...
4             and it's a really big bad thing about it
Name: clean_text, dtype: object
```

```
nlp = spacy.load("en_core_web_sm")
```

```
@spacy.Language.component("hashtag_detector")
def hashtag_detector(doc):
    hashtags = re.findall(r"#\w+", doc.text)
    doc._.hashtags = hashtags
    return doc
```

```
spacy.tokens.Doc.set_extension("hashtags", default=[])
nlp.add_pipe("hashtag_detector", last=True)

print(nlp.pipe_names)
```

```
['tok2vec', 'tagger', 'parser', 'attribute_ruler', 'lemmatizer', 'ner', 'hashtag_detector']
```

```
docs = list(nlp.pipe(data['clean_text']))
```

```
lemmas = []

for doc in docs:
    for token in doc:
        if not token.is_stop and not token.is_punct:
            lemmas.append(token.lemma_)

print(lemmas[:10])
```

```
['say', 'plus', 'add', 'commercial', 'experience', 'tacky', 'today', 'mean', 'need', 'trip']
```

```
pos_tags = []

for doc in docs:
    for token in doc:
        pos_tags.append(token.pos_)

print(pos_tags[:10])
```

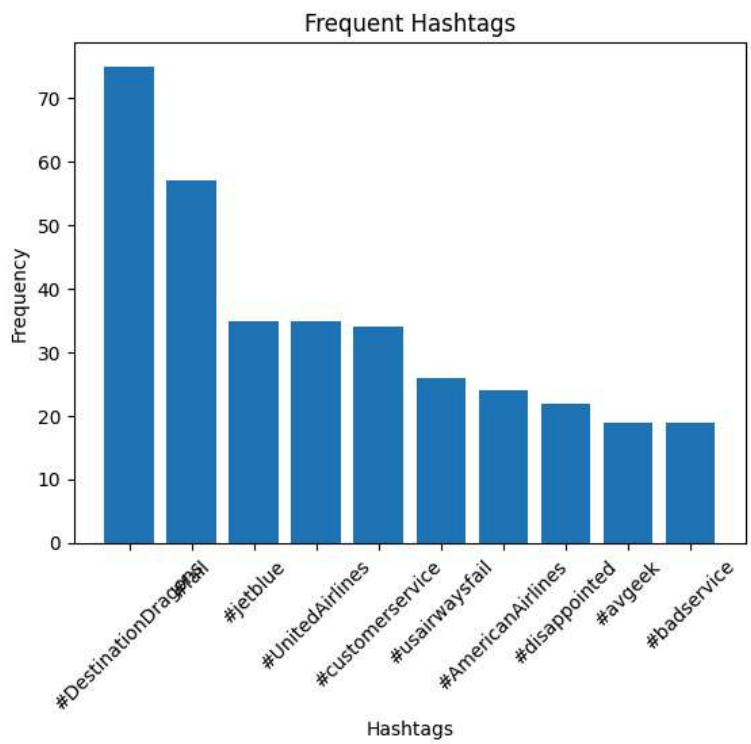
```
['PRON', 'VERB', 'PUNCT', 'CCONJ', 'PRON', 'AUX', 'VERB', 'NOUN', 'ADP', 'DET']
```

```
all_hashtags = []
```

```
for doc in docs:
    all_hashtags.extend(doc._.hashtags)

hashtag_freq = Counter(all_hashtags)
print(hashtag_freq.most_common(10))
```

```
[('#DestinationDragons', 75), ('#fail', 57), ('#jetblue', 35), ('#UnitedAirlines', 35), ('#customerservice', 34), ('#usairways
```

```
hashtags, counts = zip(*hashtag_freq.most_common(10))

plt.bar(hashtags, counts)
plt.title("Frequent Hashtags")
plt.xlabel("Hashtags")
plt.ylabel("Frequency")
plt.xticks(rotation=45)
plt.show()
```



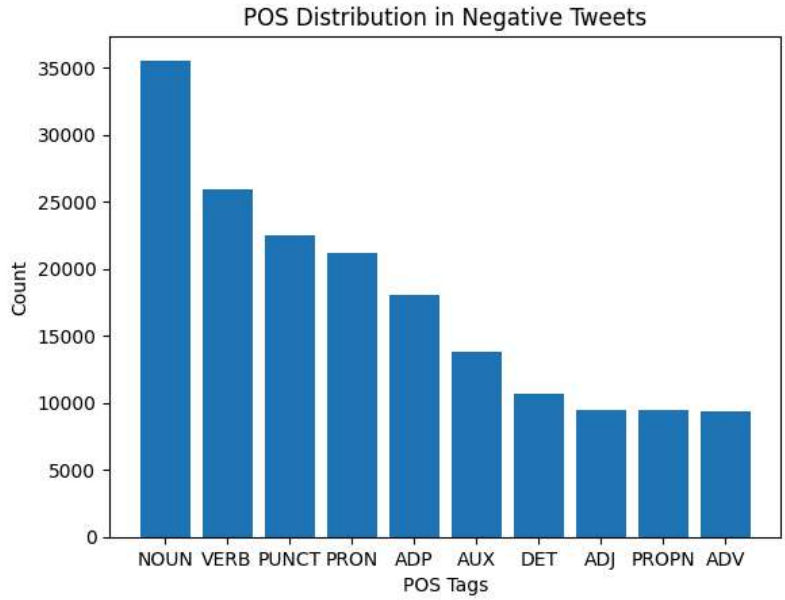

```
negative_texts = data[data['airline_sentiment'] == 'negative']['clean_text']
negative_docs = list(nlp.pipe(negative_texts))
```

```
neg_pos = []

for doc in negative_docs:
    for token in doc:
        neg_pos.append(token.pos_)

pos_freq = Counter(neg_pos)
print(pos_freq.most_common(10))
```

```
[('NOUN', 35561), ('VERB', 25959), ('PUNCT', 22490), ('PRON', 21204), ('ADP', 18095), ('AUX', 13794), ('DET', 10703), ('ADJ',
```

```
labels, values = zip(*pos_freq.most_common(10))

plt.bar(labels, values)
plt.title("POS Distribution in Negative Tweets")
plt.xlabel("POS Tags")
plt.ylabel("Count")
```

```
plt.show()
```


POS Distribution in Negative Tweets
Count
35000
30000
25000
20000
15000
10000
5000
0
NOUN
VERB
PUNCT
PRON
ADP
AUX
DET
ADJ
PROPN
ADV
POS Tags